

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 4: BOUNDARY TRACKING PROCEDURES

SLO 1: Understand and implement boundary tracking techniques to extract object contours in binary images.

THINK AND APPLY SOLUTIONS

Given Object Shapes: **Object A (Square)** and **Object B (Circle)**.

1. Which object will have a longer boundary (assuming equal area)?

Object A (Square). For a given surface area, a circle minimizes perimeter length relative to any polygon. Thus, a square enclosing the same area has a longer perimeter/boundary than a circle.

2. How can boundary tracking help in shape recognition?

Boundary tracking extracts continuous contour coordinates, allowing algorithms to derive shape descriptors such as perimeter length, curvature, compactness, perimeter-to-area ratio, and Freeman chain codes, which uniquely identify and classify distinct shapes regardless of position or scale.

MINI CHALLENGE SOLUTION

Why is boundary tracking useful in applications like object detection, medical image analysis, and character recognition?

Boundary tracking captures essential structural geometry while eliminating redundant internal data:

- **Object Detection:** Isolates object silhouettes from complex backgrounds for precise localization.
- **Medical Image Analysis:** Accurately delineates organ boundaries, tumor margins, or cell walls to evaluate shapes and measure tissue growth.
- **Character Recognition (OCR):** Extracts stroke contours and topology of alphanumeric characters for invariant classification.

REFLECTION SOLUTION

Write one advantage of extracting object contours instead of analyzing all pixels in the object.

Drastic reduction in computational complexity and memory: Boundary tracking processes only 1D contour pixel sequences instead of 2D full-region pixel matrices, significantly accelerating processing speed while preserving critical structural shape features.